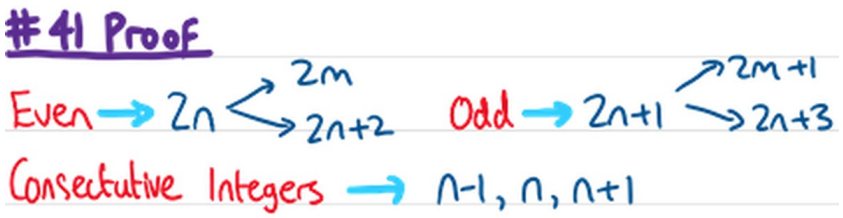
Worksheet 20 · Proof IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Use algebra to prove a given statement • Represent odd, even and consecutive numbers algebraically • Write a clear, logical proof		
Grade: / 61	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes



Questions *Proof*

Question 1 (4 marks)

Prove algebraically that the difference between the squares of any two consecutive odd numbers is always a multiple of 8.

Question 2 (4 marks)

(a) Show that $x(x - 1)(x + 1) = x^3 - x$

(1)

(b) Prove that the difference between a whole number and the cube of this number is always a multiple of 6

(3)

Question 3 (3 marks)

Using algebra, prove that, given any 3 consecutive whole numbers, the sum of the square of the smallest number and the square of the largest number is always 2 more than twice the square of the middle number.

Question 4 (32 marks)

Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

Question 5 (3 marks)

Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

Question 6 (3 marks)

Prove algebraically that, for any three consecutive even numbers,

the sum of the squares of the smallest even number and the largest even number is 8 more than twice the square of the middle even number.

Question 7 (3 marks)

Using algebra, prove that when the square of any odd number is divided by 4 the remainder is 1

Question 8 (3 marks)

Prove algebraically that the product of any two odd numbers is always an odd number.

Question 9 **(3 marks)**

N is a multiple of 5

$A = N + 1$
$B = N - 1$

Prove, using algebra, that $A^2 - B^2$ is always a multiple of 20

Question 10 **(3 marks)**

Prove algebraically that the difference between the squares of any two consecutive odd numbers is always a multiple of 8